A CMOS Compatible Energy Efficient 6T NV-SRAM-Based Accelerator Employing Binary Neural Networks in 22 nm FDSOI Technology

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***Abstract*—In this paper, a CMOS compatible non-volatile 6T-SRAM (NV-SRAM) based accelerator is proposed using a Compute-In-Memory (CIM) and Binary Neural Network (BNN) architecture to scale the persistent and energy-efficient on-chip storage for Edge AI systems. The proposed NV-SRAM facilitate reliable on-chip storage for weights and feature maps without off-chip reloading, providing data persistence and security as well as reducing information lost. As a case study, the proposed accelerator utilizes a (400 × 200) NV-SRAM array to enable binarized 20 × 20 MNIST image processing and efficient mapping of 400-bit activation vectors with the assistance of the specially designed peripheral circuits. Especially, on-chip in-memory XNOR and pop-count operations reduce inter-chip data movement, and as a result, lead to an energy-efficient processing. The design is implemented and validated in 22 nm FDSOI Technology with 0.8 V power supply to achieve throughput: 8 TOPS (8000 GOPS) and 200 TOPS/W and inference accuracy: 96.85% with MLP for MNIST, showing high energy efficiency, scalable parallelism, and small silicon area. As compared to state-of-the-art, the proposed NV-SRAM based BNN accelerator combines non-volatility, high computational throughput, and CMOS compatibility, making it a strong candidate for Internet of Things (IoT), robotics, autonomous sensors, and embedded vision in Edge AI platforms.**

***Index Terms*—Binary neural network (BNN), 6T SRAM, compute-in-memory (CIM), MNIST dataset, low-power, non-volatile SRAM (NV-SRAM), accelerator.**

I. Introduction

ARTIFICIAL intelligence (AI) systems for applications in Internet of Things (IoT), robotics, autonomous sensors, wearables, and embedded vision require hardware platforms that satisfy ultra-low energy consumption, fast inference, high reliability data security, and persistent memory operation concurrently. However, traditional computer architectures for AI accelerators have a bottleneck due to logic-memory separation. Such a memory-wall problem is the dominant term in energy consumption and latency, which results in low efficiency for Edge AI systems with limited power and resources [1].

The Binarized Neural Network (BNN) dramatically reduces computation complexity by quantizing both weights and activations into 1-bit, which substitutes the expensive Multiply-Accumulate (MAC) operation with efficient XNOR and popcount operations. These properties make BNN a highly suitable candidate for Compute-In-Memory (CIM) systems. Static Random-Access Memory (SRAM) is commonly used for the CIM because of its high speed [2], [3], [4]. But SRAM is volatile and needs sustained power to keep data or to reload off-chip weight data every power on/off cycle, etc., resulting in high dynamic power and standby leakage, long latency, and even security risks, as SRAMs have to release data to additional devices or third parties before the reloading operations. Non-Volatile Memory (NVM) technologies such as Resistive Random-Access Memory (ReRAM), Magnetoresistive Random-Access Memory (MRAM) and Phase-Change Memory (PCM) offer data persistence, but at the cost of high write energy, endurance degradation, device variability, and low compatibility with Complementary Metal Oxide Semiconductor (CMOS) techniques. To solve this problem, nonvolatile SRAM (NV-SRAM) is proposed to take advantage of the speediness of SRAM while retaining information [5], [6]. Nevertheless, there is limited work on NV-SRAM targeting CIM, and most existing BNN accelerators still adopt SRAM-based progressively-weighted-CIM arrays, where weight information will disappear during power gating/off.

Despite extensive prior research, no existing approach achieves all three: true CMOS-compatible NV-SRAM, in-memory XNOR and pop-count operations, and fully persistent BNN inference within a 6T SRAM array. This paper bridges this gap by presenting the 6T CMOS compatible NV-SRAM with in-memory XNOR and pop-count operations for BNN computation to guarantee the entire BNN inherently non-volatile at circuit, architecture, and system levels. It allows persistent weight data, instant-on inference, no off-chip weight data reloading, ultra-low standby power consumption, and data security in a single accelerator, which has not all been demonstrated so far in state-of-the-art. The main contributions of this paper are as follows. It implements: (1) a CMOS-compatible 6T NV-SRAM with true in-memory BNN computation; (2) a

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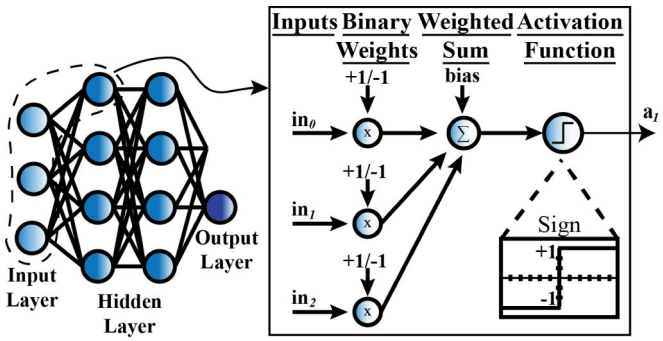


Fig. 1. Binarized Neural Networks [9].

fully persistent BNN accelerator with zero-waiting instant-on operations and ultra-low standby leakage; (3) in-memory XNOR and pop-count computation to achieve ultra-low energy efficiency of 5 fJ/op; and (4) a validation based on 22 nm FDSOI Technology.

This paper is organized in the following sections. Section II introduces the background and the related work. The proposed circuit and architecture for NV-SRAM based BNN accelerator is described in Section III. Section IV shows the simulation results. Comparisons are listed in section V. The paper is concluded in Section VI.

II. Background and Related Work

A. BNNs and Hardware Acceleration

BNN is one of the most efficient families in the realm of neural networks, where weights and activations are both binarized, i.e., quantized to +1 and −1. This quantization significantly lowers the memory overhead and reduces computation complexity, well-suited for fast and energy-efficient inference on Edge devices, e.g., IoT nodes, robotics, and wearables [7], since floating-point multiply-and-accumulate operations, upon which conventional Deep Neural Networks (DNNs) are based, are costly in terms of power and demand high memory bandwidth. On the other hand, BNNs, shown as Fig. 1, have employed bitwise XNOR followed by pop-count to compute dot products realizing extremely parallel computing in memory arrays. These qualities render BNNs particularly attractive for Edge AI use cases, where power and area constraints are paramount, while accuracies can be sacrificed to some extent [8].

B. SRAM for Compute-In-Memory (CIM) With BNNs

Several prior works aim to design SRAM-based accelerators that effectively exploit the computational advantages of BNNs. 6T SRAM arrays have traditionally been tuned for CIM, where the XNOR and pop-count operations are performed directly along bit-lines to minimize off-chip memory accesses and increase throughput [10]. Redundant cell nodes, such as 8T and 10T SRAM cells to improve read stability, reliability, and noise margin, but they come at the cost of area and energy. XNOR-SRAM and charge-domain SRAM macros achieve efficient computation of binary convolutions and fully connected layers, confirming the validity of the

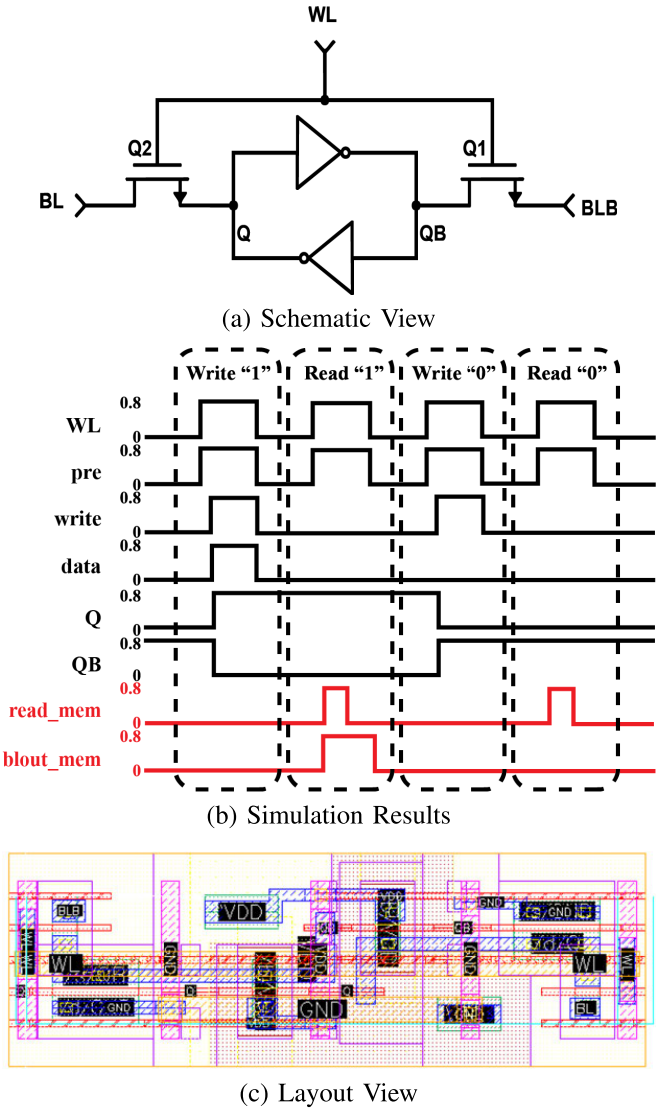


Fig. 2. A Conventional 6T SRAM Bit-cell.

concept SRAM-CIM in BNN workloads. Other than SRAM, new Non-Volatile Memories (NVMs) including ReRAM [11], [12], [13], MRAM, and PCM have been examined for BNN acceleration due to their non-volatile nature. Such designs have instant-on features and low leakage energy, but suffer from high integration complexity, low write endurance, and high write energy that preclude them being used widely in CMOS-compatible Edge AI accelerators. Although SRAM-CIM is energy-efficient computation, the volatility and nonexistence of permanent storage is still one of the serious constraints for secure and robust BNN implementations.

C. Problems With Conventional Designs

From the background of the current art, three major challenges are observed in SRAM-based BNN accelerators. First, volatility of conventional 6T SRAM cells shown in Fig. 2 need weight reloading from external memory when power off/on cycle, inducing high-energy, latency, etc. penalties. Second, leakage currents from the SRAM array generate static

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power consumption that constrains energy-efficiency for low-power Edge devices [6], [14]. Third, the weights and feature maps are transferred off-chip, which sacrifices data security such that these architectures are susceptible to applications demanding protection against illegal access and intermittent power supply. In Fig. 2 (b), when the memory mode is on and "read_mem" is activated, then the memory can be read through "blout_mem". Traditional NVMs exist with retention capability, but are typically not as efficient in terms of area, speed, or CMOS compatibility, and can not benefit from a trade-off among efficiency, security, and manufacturability. In response to the drawbacks of traditional SRAM reloading and NVM endurance problems, in this work, a new CMOS-compatible 6T NV-SRAM cell based BNN accelerator is proposed. The NV-SRAM stores weights and feature maps without reloading data, allowing instant-on operation with lower leakage and increased security, whereas the compact 6T architecture remains area-efficient and manufacturable in 22 nm FDSOI Technology. The input vectors and the weights are stored and computed directly into the NV-SRAM array. Especially, the local XNOR-popcount reduces off-chip data transfer and enables high parallelism with high power efficiency.

D. Novelty in Proposed Design

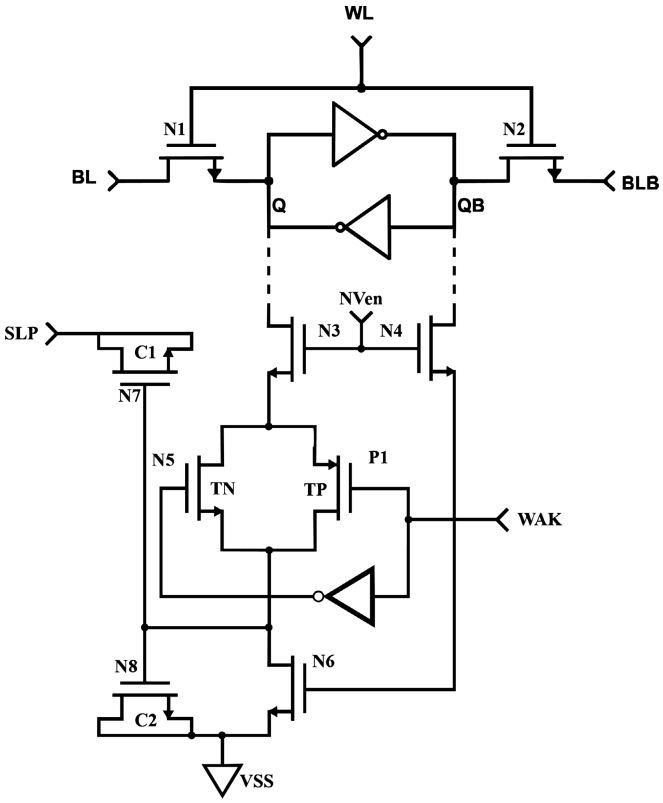
Novelties of the proposed design include: (1) unlike prior NV-BNN accelerators [5], [6], [15], [16], [17], [18], [19] that utilize SRAM+MRAM, SRAM+RRAM, SRAM+PCM, or Fowler-Nordheim tunneling mechanism, this work uses Extended Gate Field Effect Transistor (EGFET) as coupling capacitors for true non-volatility without Static Noise Margin (SNM) penalty, eliminating NVM–SRAM data movement and enabling ultra-low energy operation (~5 fJ/op), especially with CMOS compatibility; (2) different with previous designs [20], [21], it avoids the complex analog circuits such as Analog-to-Digital Converter (ADC) and employs fully digital XNOR/pop-count CIM inside a single 6T SRAM cell to simplify the design methodology and improve operation reliability; (3) Because of EGFET components, the programming voltage is reduced from over 6 V [5] to 1.9 V, resulting in further increased energy-efficiency and smaller programming latency; and (4) a new methodology is proposed for a scalability and flexibility design to adapt from the case study to multi-layer or deep BNN architectures.

III. Proposed Design

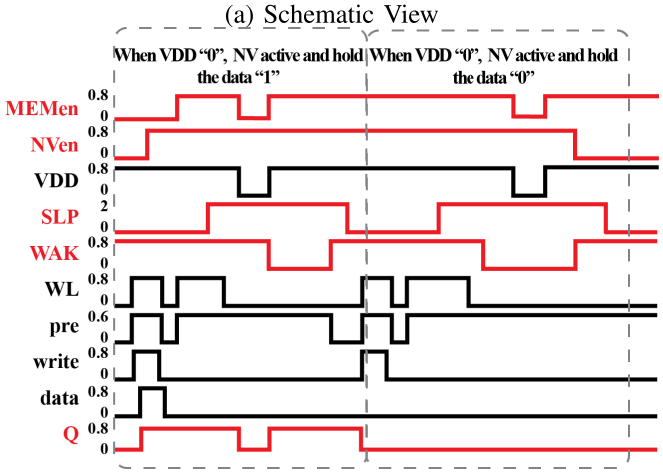
A. Non-Volatile 6T SRAM

Traditional SRAM cells shown in Fig. 2 are volatile, meaning that their content is lost upon disconnecting the power. To solve this problem, NV-SRAM architectures entrapping the speed and resilience of SRAM with non-volatile data retention have been proposed. One example is the capacitor-coupled non-volatile SRAM (NV-SRAM) with extra capacitive devices and control transistors to a normal SRAM structure using Fowler-Nordheim tunneling mechanism [5], [6].

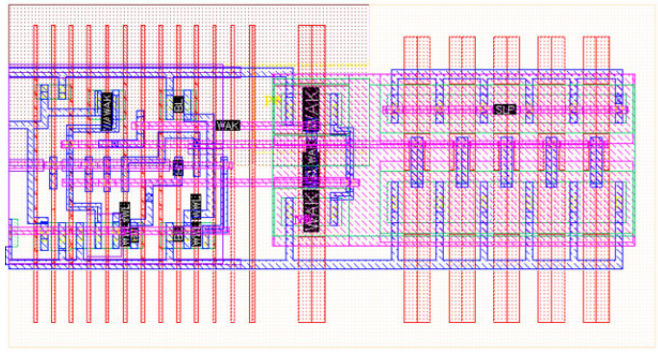
Different with [5], [6], our proposed NV-SRAM cell, shown in Fig. 3 (a), is adopted from a 6T-SRAM cell that is enhanced with two capacitors and a transmission gate. The coupling capacitors C1 and C2 are realized by Extended Gate



(a) Schematic View



(b) Simulation Results



(c) Layout of NV-SRAM Cell

Fig. 3. A Non-Volatile 6T SRAM Bit-cell.

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Field Effect Transistor (EGFET) to act as the non-volatile storage element. A restoring path is enabled and disabled by a transmission gate that is made of the transistors TN and TP in response to system control signals.

As shown in Fig. 3 (b), in the normal operation mode ("WAK" is "1" and "SLP" is "0"), the transmission gate is off, so that the NV-SRAM cell operates the same as a normal 6T SRAM cell, thereby enabling fast access time, high Static Noise Margin (SNM), and robustness with regard to PVT variations. In the Sleep mode, a low "WAK" is given for a short time and an active high "SLP" pulse is given by a charge pump. At this point, the "SLP" voltage allows charging of capacitor C1 made of the EGFET. Depending on how data is stored, electrons are either ejected or passed to the C1 [22]. When the data of the internal node Q is "1", the corresponding complement node QB is "0", and hence the C1 preserves a high voltage, which represents a non-volatile "1". On the other hand, when Q is "0", C1 discharges the voltage to the ground and stores a non-volatile "0". In this backup process, the supply voltage enables the memory to rest in ultra-low leakage but retains data. When the potential of C1 is high, the node Q is pulled up to "1" and, conversely, when the potential of C1 is low, the node Q is driven to "0" to recover the original logic state of the SRAM cell. This process provides a mean to assert data recovery after a power cycle that can retain data non-volatility in a flash-like memory while still offering the high-speed access of SRAM [5], [15]. Backup and restoration also need to be performed at a proper timing to reduce performance overhead. The backup time is determined based on the response of the charge pump and, therefore, an actual sleep time, namely, T_{sleep}, is different from the standby time, $T_{standby}$ of the memory. This connection can be formulated as:

$$T_{sleep} = T_{standby} - T_{cp}$$

where T_{cp} is the charge pump feedback time. Energy-efficiency is benefited by standby longer than the charge pump delay for low-leakage operation. Additionally, the recovery time should still be brief in order not to degrade performance when functioning is resumed [23]. Circuits based on 22 nm FDSOI Technology with a 0.8 V supply voltage is simulated and shows that reliable backup and restore for both "1" and "0" states can be achieved. Simulation waveforms and layout of NV-SRAM cell are shown in Figs. 3 (b) and (c). In Fig. 3 (b), it can be seen that when the system is in Non volatile mode (NVen) and "SLP" is activated, it keeps data which is reflecting in Q. This CMOS compatible NV-SRAM validates ultra-low leakage energy when the power is off while maintaining strong performance during active mode, which makes it a non-volatile memory alternative for low-power embedded and Edge AI systems that are demanded to take retention [24].

The robustness of the 6T NV-SRAM cell is examined in read, write and hold SNM shown in Fig. 4. SNM is extracted from the butterfly curve as the largest square that entirely fits within the voltage transfer characteristics (VTCs) of the cross-coupled inverters. The SNM is defined by the side length of the largest square that can be fitted between the inverter characteristics [25], [26].

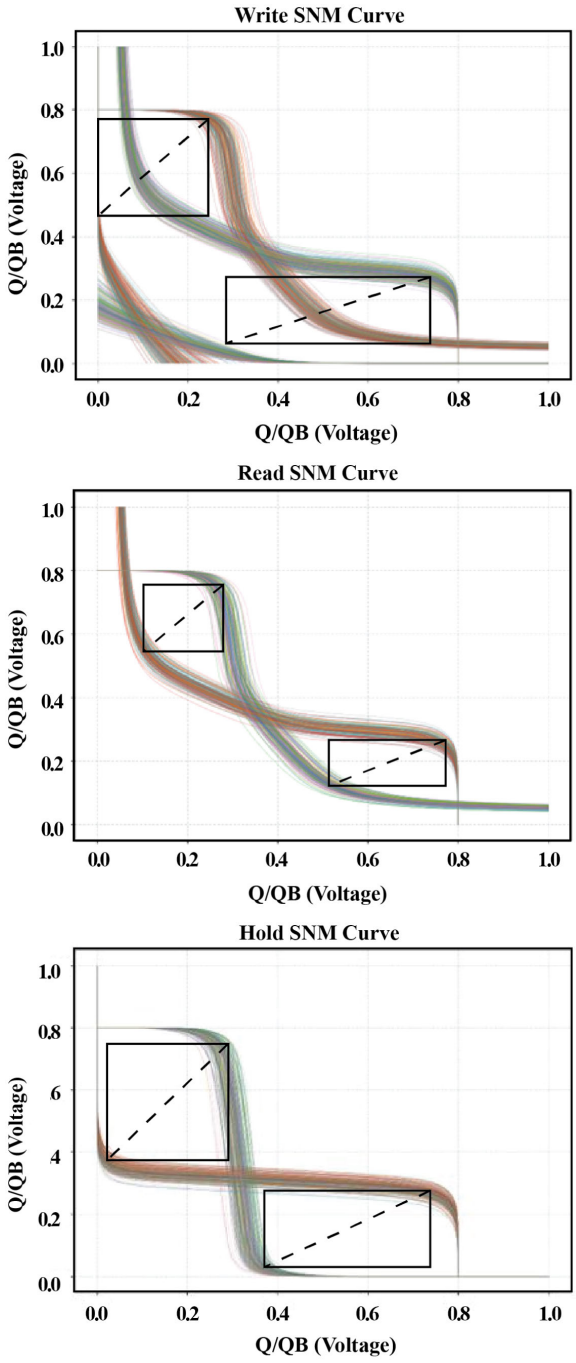


Fig. 4. Butterfly Curves of NV-SRAM (Read, Write and Hold SNM).

Reading activates a connection of bit lines to the storage nodes Q/QB, potentially decreasing the SNM in traditional SRAM. In the case of the NV-SRAM case, the extra capacitors and control transistors all remain electrically isolated.

During the write operation, the bit-lines dominate the cross-coupled inverters, and so the NV path remains disabled. Then, express SNM is maintained which depends primarily on the cell ratio and pull-up ratio. In hold mode, when WL is "0", SNM depends on inverter scaling only. Additional, capacitance by the NV elements does not have any relevant impact. Simulation results show that the proposed NV-SRAM has not much degraded SNM relative to the typical 6T-SRAM.

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TABLE I
INPUT WEIGHT CONFIGURATION FOR IN-MEMORY XNOR

Input	Weight	WL	WLB	Q	QB	V_{SUM}	Output
1	1	VDD	GND	VDD	GND	↑ (PU)	1
1	−1	VDD	GND	GND	VDD	↓ (PD)	−1
−1	1	GND	VDD	VDD	GND	↓ (PD)	−1
−1	−1	GND	VDD	GND	VDD	↑ (PU)	1

Monte Carlo Simulation results shown in Fig. 4 validate these analyses. For 22 nm FDSOI Technology at $VDD = 0.8\ V$, the 6T NV-SRAM proposed achieves a hold SNM of about 210 mV, which is less than 2% of the baseline 6T SRAM cell. Furthermore, the read SNM is also maintained which is 140 mV, and the write SNM is retained over supply voltage scaling which is 100 mV. This shows that non-volatility integration does not degrade cell endurance but allows robust backup and recovery after power-cycling.

B. Non-Volatile 6T SRAM-Based Binary Neural Network

BNNs involve weights and activations as binary (1-bit) values requiring highly energy-efficient storage and computation. Being quick to access and suitable for CIM designs makes 6T SRAM cells suitable for BNN accelerators, where bitwise operations (like XNOR and pop-count) are carried out directly on the content of the memory array. But conventional SRAM is volatile, meaning it cannot hold data without power. To address this bottleneck, 6T NV-SRAM introduced in III-A incorporates auxiliary structures to make the data non-volatile even under power gating/off with ultra-low leakage standby modes and instant data recovery upon wake-up.

Table I corresponds to the operating rule of the SRAM-based CIM bit-cell, in which the input and the weight are encoded by the bipolar format [27], i.e. +1 and −1. The operation of the multiplication $output = input \times weight$ is depicted, which clearly demonstrates how the SRAM word line signals (WL and WLB) with the cell internal storage nodes (Q and QB) interact to dominate the resulting sum current V_{SUM}. In essence, Fig. 5 shows how the hardware implements the multiplication of an input by a stored weight in the circuit. WL and WLB is splited as it is seen in the Fig. 5. And because of the combination summerised in the Table I, "blout_CIM" is working when "read_CIM" signal is activated, as shown in Fig. 5 for simulation results.

For the input 1 and weight 1, 1 causes word line (WL) to go logic high value (VDD) and WLB to go logic low value (GND). Because the charged weight at $Q = VDD$ and $QB = GND$. Get pull-up (PU) path on, and the increase on the summation node ($V_{SUM} \uparrow$) is obtained. This yields an output of +1, which is indeed the correct product for $1 \times 1 = 1$.

For the input 1 and weight −1, WL is VDD again with WLB as GND and the weight stored is inverted ($Q = GND, QB = VDD$). In the meantime, the pull-down (PD) path is enabled, and the summation line discharges downwardly ($V_{SUM} \downarrow$). This result is −1, which matches the result of the multiplication $1 \times (-1) = -1$.

For the input −1 and weight 1, WL is GND and WLB is VDD. By storing $Q = VDD$ and $QB = GND$ for $weight = 1$, the access transistors short the summation line to a pull-down (PD) path, and V_{SUM} charges. This yields an answer of −1 as the output as well, which once more aligns with the desired multiplication, $(-1) \times 1 = -1$.

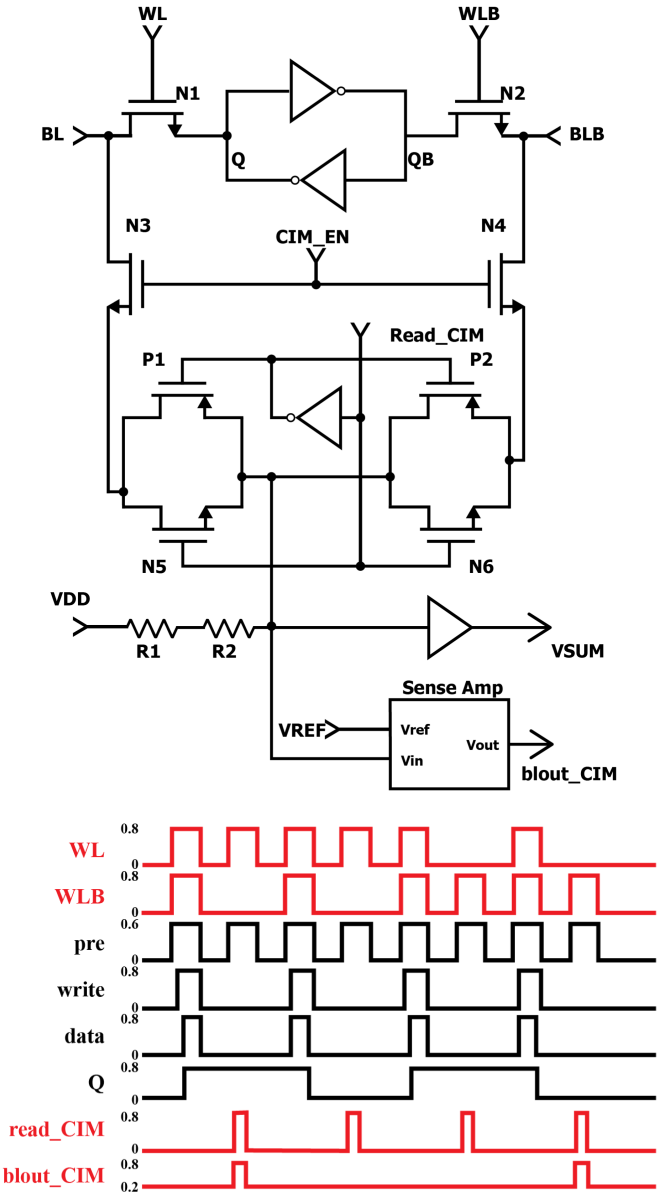


Fig. 5. SRAM-based CIM Circuitry (Circuitry and Simulation Waveform of XNOR Operation).

In addition, for the input −1 and weight −1, the WL is GND and the WLB is VDD, and the stored weight in the SRAM is $Q = GND$ and $QB = VDD$. In this setting, pull-up (PU) path is on, therefore contributes high ($V_{SUM} \uparrow$) This multiplication gives an outcome of +1, which is in agreement with $(-1) \times (-1) = +1$.

It can be concluded from the Table I that the SRAM bit-cell naturally realizes a multiplication between input and weight values that are encoded as ±1. The storage nodes (Q, QB) determine the weight that is stored, while WL, WLB determine the response to the applied input. The summation node is pulled up or pulled down, depending on which path

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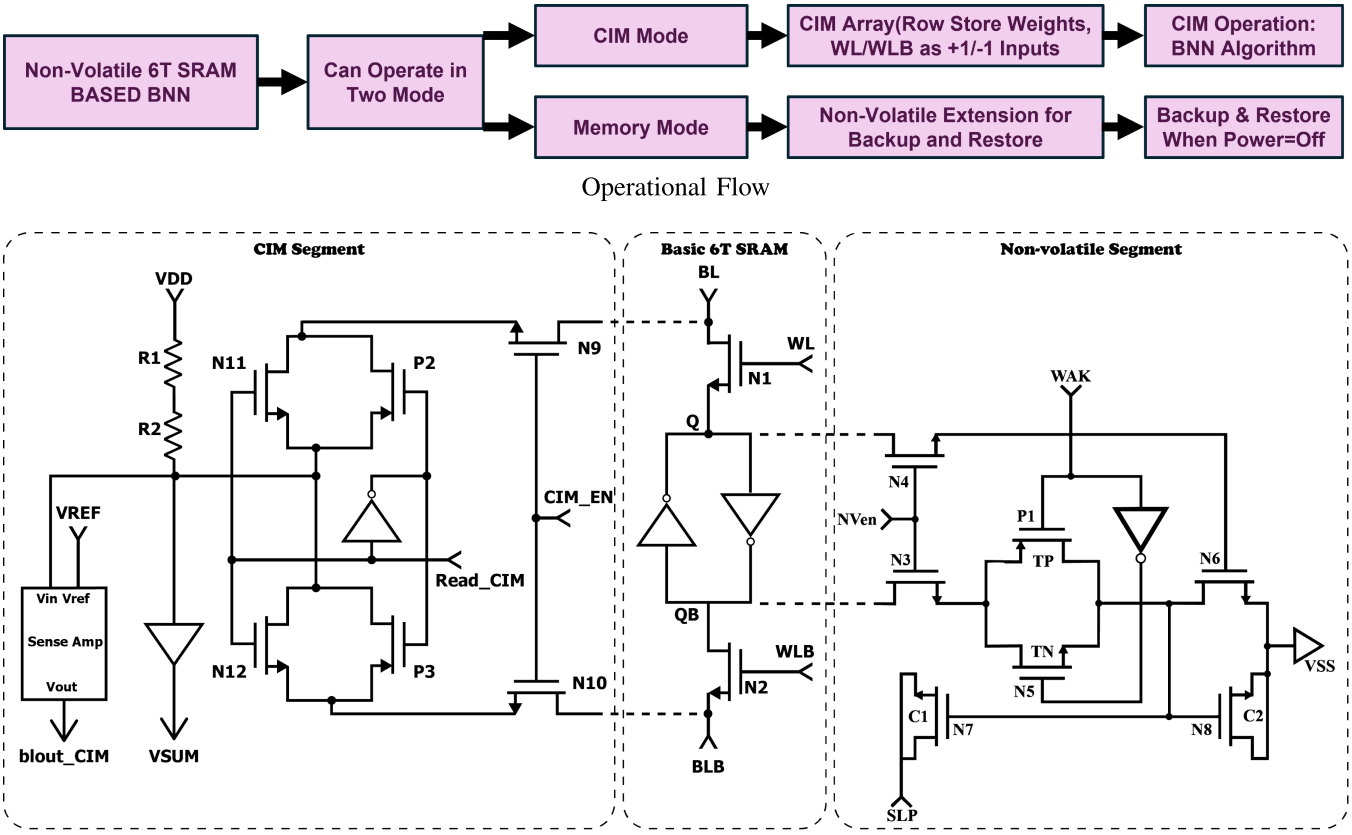


Fig. 6. Operational Flow and Circuit Design of NV-SRAM Based BNN Architecture.

is dominant, and the correct sign of the output in binary multiplication is obtained.

C. Binary Neural Network Deployment

A 6T NV-SRAM cell is used to the BNN, it can operate as standard memory mode or CIM mode. Fig. 6 shows the flow chart and circuit design of NV-SRAM based BNN architecture. This architecture supports conventional read and write operations, and CIM. In standard memory mode, the SRAM functions as a standard SRAM cell. Data can be read or written using bit lines (BL and BLB), and sense amplifiers used to sense and amplify the stored digital data [28]. Additional transistors and capacitors are added for making the structure having non-volatile storage capability. During sleep or power-down periods, the stored logic state ("1" or "0") is backed up to a dedicated non-volatile storage node. A plurality of NV-SRAM cells is grouped to constitute an NV-SRAM array at the array level. Each row of the array can be binary weights that are input to a neural network [29]. Divided word line control (WL/WLB) permits the inputs to be presented as +1 or −1 actively. Bit-line currents are summed up to produce an analog dot product output V_{SUM}. This design is successful in the sense that storage and computation are fused together by reducing high overheads to data movement.

In the CIM mode, the input activations are fed through the word lines. These inputs combine with the stored weights and perform an XNOR operation. Matching input and memory weights (e.g., Input equal to "1", Stored equal to "1") will produce a non-zero match response, while other input and stored weight will produce a zero contribution [30]. Several rows are actuated at once and the sum of the pull-up and pull-down currents are integrated in V_{SUM}. A sense amp compares the V_{SUM} with a reference supply voltage V_{ref}, to create the final binary neuron output.

Fig. 7 presents a case study for a hardware implementation of NV-SRAM based BNN architecture where the Input Representation: A 20×20 MNIST image is binarized into a 400-bit activation vector. They are stored in 6T NV-SRAM cells.

Layer Computation:

Input Layer: 20×20 *MNIST image* $\rightarrow$ 400 *pixels* $\rightarrow$ represents 400-*bit* binary activation vector [31].

Hidden Layer: 400 *inputs* $\times$ 200 *hidden neurons* $= 80,000$ binary weights in the NV-SRAM. In CIM mode, hidden activations are computed using *XNOR+pop-count* operations.

Output Layer: The *XNOR+pop-count* is used in the same manner to generate the 10-class digit classification output. 200 *hidden neurons* $\times$ 10 *output neurons* $= 2,000$ binary weights.

NV-SRAM maintains trained weights and intermediate data during sleep, also ultimately minimize standby leakage and fast wake-up time [32].

Connection Logic:

Input→ Hidden Layer: Input activation vector is generated firstly. Each of the 400 input bits will interact with all 200 hidden neurons. This input vector is fed to 400 × 200 NV-SRAM array in CIM fashion [33].

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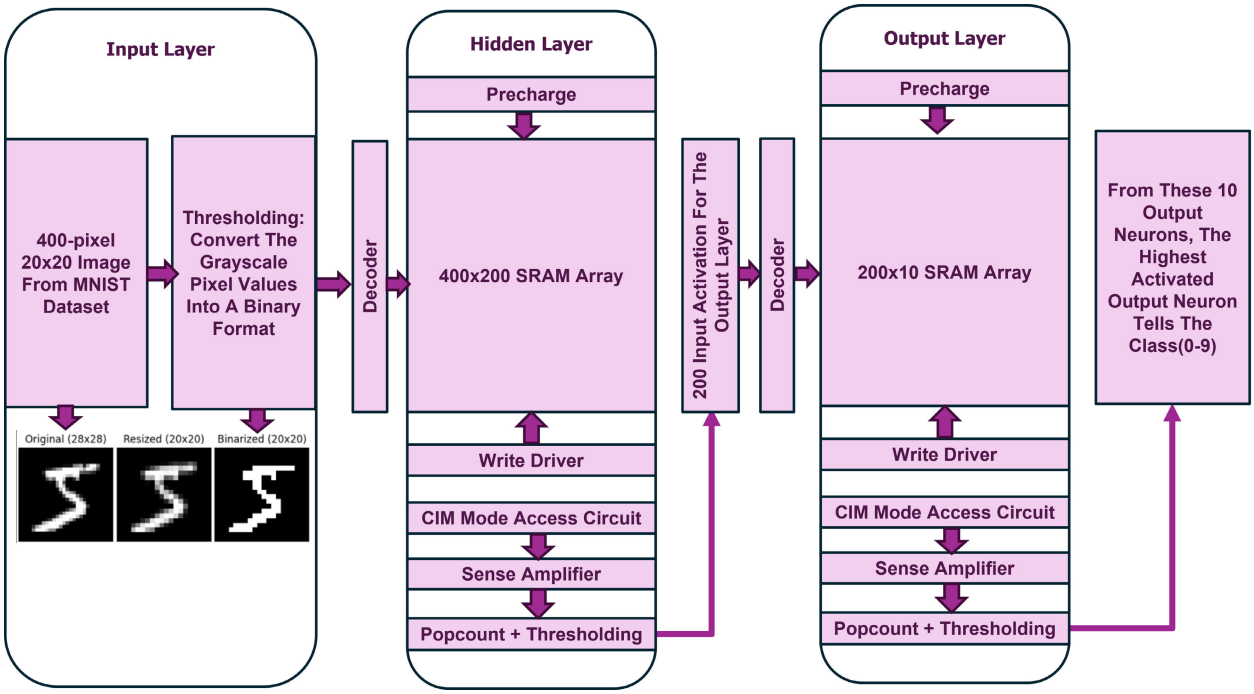


Fig. 7. Hardware Implementation of NV-SRAM Based BNN Architecture.

A binarized image is a 20×20 binary matrix, with elements:

$$x_{i,j} = \begin{cases} 1, & \text{if pixel value} > 127 \\ 0, & \text{if pixel value} \le 127 \end{cases}$$

Flatten it to a vector of length 400 (for 20×20 image):

$$\mathbf{x} = [x_1, x_2, \ldots, x_{400}]^T \in \{0,1\}^{400\times1}$$

Equation for word line activation:

$$WL_i = \begin{cases} 1, & \text{if } x_i = 1 \\ 0, & \text{if } x_i = 0 \end{cases}$$

Weights (Stored in NV-SRAM Cells via Write Driver): In a BNN, the weights are binary $0, 1$ or $-1, +1$ stored in NV-SRAM. First layer has 400 inputs $\rightarrow$ 200 hidden neurons, then the weight matrix is:

$$W = \begin{bmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,200} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,200} \\ \vdots & \vdots & \ddots & \vdots \\ w_{400,1} & w_{400,2} & \cdots & w_{400,200} \end{bmatrix} \in \{0,1\}^{400\times200}$$

Each column corresponds to the weights of one neuron. Each row corresponds to a word line row in the NV-SRAM array. During weight programming (write), the write driver sets each NV-SRAM cell according to the corresponding weight:

$$\text{SRAM}_{i,j} = \begin{cases} 1, & \text{if } w_{i,j} = 1 \\ 0, & \text{if } w_{i,j} = 0 \end{cases}$$

BNN Computation (XNOR + Popcount): Once the input activations and weights are in the array: XNOR operation between input bit x_i and stored weight $w_{(i,j)}$ for neuron j:

$$y_{i,j} = \text{XNOR}(x_i, w_{i,j}) = \begin{cases} 1, & \text{if } x_i = w_{i,j} \\ 0, & \text{if } x_i \neq w_{i,j} \end{cases}$$

Pop count/summation over inputs to get neuron activation:

$$a_j = \sum_{i=1}^{400} y_{i,j}$$

where a_j is the pre-activation of neuron j. This is the CIM result, which is read from bit lines via sense amplifiers and then pop count is done.

Hidden Layer$\rightarrow$ Output Layer: The 200 hidden activations produced by the first NV-SRAM array is then used as the input to the 200×10 NV-SRAM array. 200 outputs from the first array are the inputs of the second array. From Hidden Layer to Output Layer, the 400×200 NV-SRAM array produces 200 hidden neuron activations:

$$h_1, h_2, h_3, h_4, h_5, \ldots, h_{200}$$

These are binary activations $(0/1)$ or $(-1/+1)$.

The 200×10 NV-SRAM array stores weights connecting the 200 hidden neurons to the 10 output neurons (digits 0-9). So now, the 200 hidden activations become the input vector to the output layer NV-SRAM.

Output Layer: Each hidden neuron's output bit acts as an input bit for all 10 output neurons. Output neuron activation is computed same way as hidden layer: XNOR with stored weights$\rightarrow$pop count$\rightarrow$threshold.

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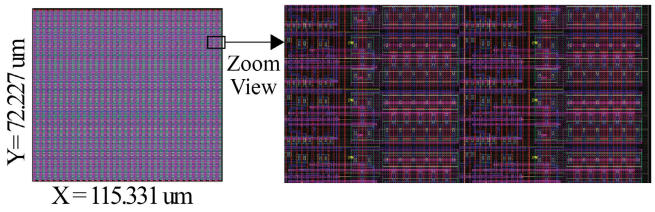


Fig. 8. Layout Design for 1KB NV-SRAM Bank.

Each of the 10 output neurons has 200 weights, stored in NV-SRAM. For each output neuron j:

$$y_j = \text{Sign}\left(\sum_{i=1}^{200} h_{i,j} \odot w_{i,j}\right)$$

Thus, 10 final outputs are obtained–one for each possible digit.

A layout design for 1KB NV-SRAM bank shown in Fig. 8 is constructed symmetrically to meet the schematic design of the NV-SRAM based BNN architecture. Symmetrical routing helps to boost the performance of the cell. The bit lines (BL/BLB) and Word lines (WL/WLB) are kept orthogonally. In addition, to get a compact design, the design approach is considered sharing nodes such as VSS, WL/WLB, VDD and BL/BLB with neighboring memory cells so that the area it took is approximately 8330 um^2 for one bank. Because totally, 82,000 weights are needed in hidden and output layers, ~10 banks will be included in the accelerator.

D. Scalability Analysis for Deeper BNN Implementations

The proposed NV-SRAM based CIM architecture is inherently scalable to deeper BNN models through modular replication of NV-SRAM arrays and hierarchical inter-layer data flow organization. While this work demonstrates a three-layer BNN (400–200–10) targeting the MNIST dataset, the same architectural principle directly extends to multi-hidden-layer BNNs without requiring any modification to the NV-SRAM cell or CIM circuitry.

Hidden Layer Scalability: For a general deep BNN with L hidden layers, the network can be expressed as:

$$N_0 \rightarrow N_1 \rightarrow N_2 \rightarrow \cdots \rightarrow N_L \rightarrow N_{\text{out}}$$

where each layer-to-layer weight matrix ($N_i \times N_{i+1}$) is mapped onto one dedicated NV-SRAM CIM array. Each array performs local XNOR and pop-count operations, and the resulting binary activations are directly forwarded to the next array without off-chip memory access. Therefore, memory and computation remain fully collocated at all depths, preventing the memory bottleneck that typically limits deep-network accelerators.

Memory Size Scalability: The total number of binary weights stored in NV-SRAM for a deep BNN scales linearly with the number of layers and neurons:

$$\text{Total Weights} = \sum_{i=0}^{L} \left(N_i \times N_{i+1}\right)$$

Since each weight is stored as one bit, the memory footprint remains extremely compact even for deeper networks.

Compute and Throughput Scalability: Each NV-SRAM array exploits massive bit-level parallelism along the word-lines and bit-lines. For layer i, all $N_i \times N_{i+1}$ XNOR and pop-count operations are executed in parallel within the memory array. Thus, the total number of operations per inference scales as:

$$\text{Operations per Inference} = \sum_{i=0}^{L} \left(N_i \times N_{i+1}\right)$$

However, because each layer operates in a highly parallel CIM fashion, the overall inference latency is primarily governed by the largest layer rather than by the total sum across all layers. This ensures that high throughput is preserved even as network depth increases.

Energy Scalability: Since all operations are performed in situ within the NV-SRAM, inference energy scales proportionally with the number of binary operations:

$$E_{\text{inference}} \approx E_{\text{op}} \times \sum_{i=0}^{L} \left(N_i \times N_{i+1}\right)$$

where $E_{\text{op}} \approx 5$ fJ/op in the proposed design. Because expensive off-chip data movement and reloading operations are completely eliminated, the proposed architecture maintains ultra-low inference energy even for deeper BNNs.

Scale to ImageNet and CIFAR-10: Based on above scalability theory for hidden layer, memory size, throughput, and energy, the proposed design methodology can be applied to various neural networks without any architectural change. Its generality follows the fact that all operations are represented in terms of binary XNOR and pop-count primitives, intricately mapped to the NV-SRAM bit-cells, thus being independent of network topology, layer-depth, or input dimensionality. As validated by the extended benchmarks reported in Table II, the same principle of NV-SRAM CIM is initially proposed for a 3-layer MLP on MNIST (400–200–10) and can be systematically extended to ResNet-18 and Vision Transformers (ViT), to support both CIFAR-10 and ImageNet-1K with input dimensions varying from 3,072 bits to 150,528 bits as well as weight storage larger than 22 million of parameters. In addition, as ResNet-18 and ViT for both CIFAR-10 and ImageNet-1K do not have any extra numerical approximation other than the typical BNN binarization, computational correctness is preserved [34], [35]. Therefore, they keep accepted inference accuracy. While the original demonstration achieves a 96.85% accuracy for MLP with MNIST on a 400×200-size NV-SRAM array (requiring ~10 KB), the scale up of such an architecture to large vision models is established to reach 92.31% and 95.0% accuracy using ResNet-18 and ViT on CIFAR-10, as well as achieve accuracies of 65.5% (ResNet-18) and 70.1% (ViT) when for ImageNet-1K. These findings validate the proposed methodology for NV-SRAM BNN accelerator have great scalability and flexibility.

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TABLE II
PROJECTED CIFAR-10 MAPPING ON PROPOSED NV-SRAM CIM ARCHITECTURE

Parameter	MLP for MNIST	ResNet-18 for ImageNet-1K [34]	ResNet-18 for CIFAR-10 [34]	ViT for ImageNet-1K [35]	ViT for CIFAR-10 [35]
Input Size	$20 \times 20 = 400$ bits	$224 \times 224 \times 3 =$ 150,528 bits	$32 \times 32 \times 3 = 3,072$ bits	$224 \times 224 \times 3 =$ 150,528 bits	$32 \times 32 \times 3 = 3,072$ bits
No. of Layers	3	18	18	12	12
Hidden Neurons / Dimension	200	4,800	4,800	384	384
Output Classes	10	1,000	10	1,000	10
Weight Count	82 K	11.2 M	11.2 M	22.1 M	22.1 M
NV-SRAM Required	~10 KB	1.4 MB	1.4 MB	2.8 MB	2.8 MB
Accuracy (%)	96.85	65.5	92.31	70.1	95.0

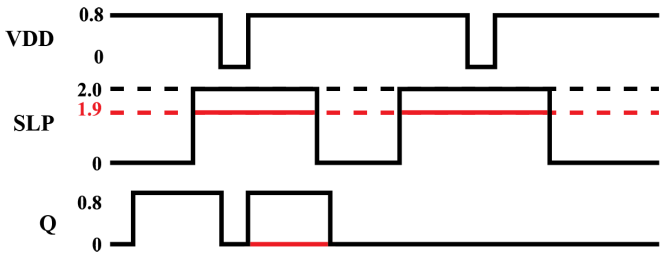


Fig. 9. Store and Restore Waveforms of Proposed 6T NV-SRAM.

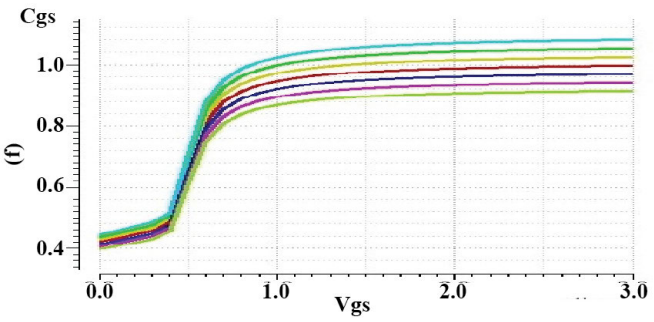


Fig. 10. Gate to Source Capacitance (C_{gs}) vs Gate to Source Voltage (V_{gs}) Characteristic of the Mosfet Based Capacitor C1.

IV. SIMULATION RESULTS

A. Transient Behavior of Propose NV-SRAM

The waveform shown in Fig. 9 is the store-restore action of the proposed 6T NV-SRAM with power on or off. In the top panel, supply voltage (VDD) is pulsed from 800 mV to 0 V to simulate power off events. The middle panel illustrates "SLP" rises to 1.9 V and trigger the non-volatile element to latch the stored state. In the lower panel, the storage node (Q) holds the voltage (~ 800 mV) during power interruption and recovers properly when VDD is re-applied, indicating stable non-volatile retention.

Fig. 10 shows the non-volatile element which is capacitor made of EGFET to source capacitance C_{gs} as a function of gate to source voltage V_{gs}. With low V_{gs} (below threshold), the device is in off-state and C_{gs} is low and constant as a result of overlap capacitance. When V_{gs} becomes closer to the threshold voltage, C_{gs} increases abruptly in accordance with the inversion layer (channel) formation. Above threshold (strong inversion), C_{gs} saturates and slightly depends on device parameters such as width and length. In the proposed design, the width of the capacitor made of transistor, C1, is kept 810 nm and gate length 20 nm.

B. Monte Carlo Results for Write, Read and Hold in NV-SRAM

In NV-SRAM implementation, it is important to have hold, write, and read reliability under PVT (Process, Voltage, Temperature) variations. Fig. 11 shows the histogram of 200 Monte Carlo samples for the hold, write and read operation is plotted [36]. The mean value of the Hold Monte Carlo simulation above is roughly 789.8 mV with an incredibly small standard deviation (1 mV), meaning that the state held by the cell is maintained with extraordinary consistency, a critical characteristic for reliable non-volatile data retention. The write Monte Carlo simulation frame is centered (median) at around 267 mV with a modest spread (54 mV), meaning writes are behaving quite reliable and constant. However, there are outliers towards high values (right tail) suggesting that a few writes take more effort to store data. On the other hand, the Monte Carlo simulation graph for reading indicates a higher mean of 407.2 mV and wider spread (~ 87.8 mV) as well as left skew indicating reads are more variable and sensitive to mismatches [37]. All together, these data demonstrate a typical NV-SRAM trade-off; hold stability is best, writes are comparable, but reads are more sensitive to variability, potentially affecting sensing reliability in large NV-SRAM arrays.

C. Results of NV-SRAM Based BNN Architecture and CIM Mode With Power-On

The waveform shown in Fig. 12 represents the transient simulation of a 6T NV-SRAM cell integrated with a sense amplifier, designed for BNN operations. The signals illustrate the different phases of memory operation-write, read, and hold along with sense amplifier activation.

The word line (WL) and its complement WLB control the access transistors of the NV-SRAM cell. When WL is asserted high, the memory cell is accessible for either read or write operations. The pre-charge signal "pre" equalizes the bit lines (BL and BLB) before each read, ensuring proper sensing. During write, the write signal enables data storage, where the input drives the value onto the bit lines, which is then latched into the storage node Q. Once stored, Q maintains its value during hold.

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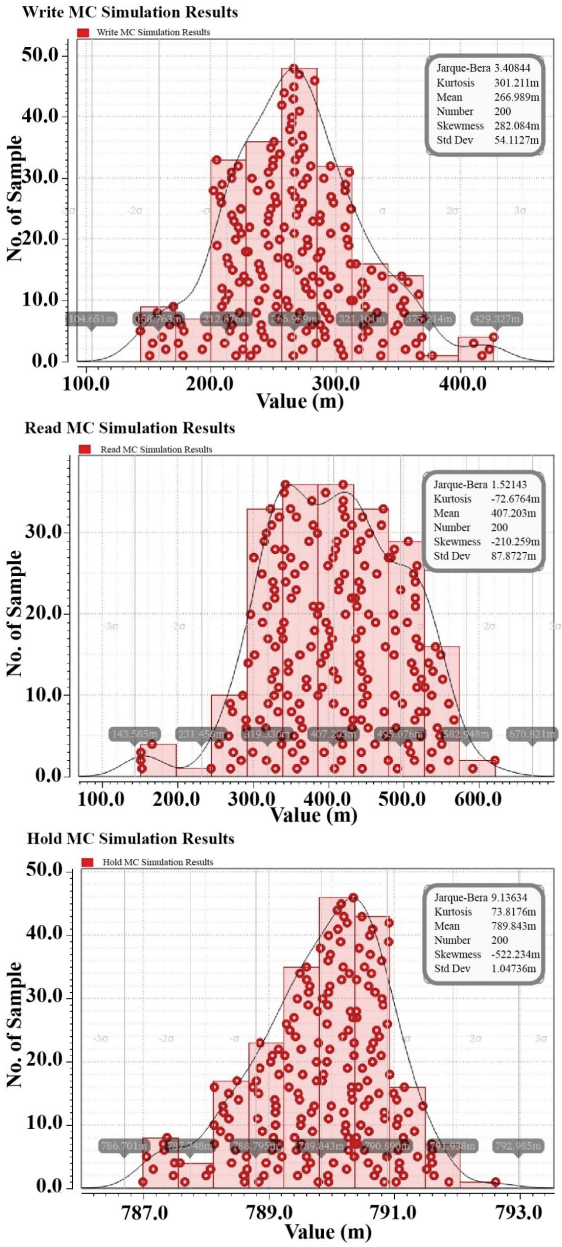


Fig. 11. Monte Carlo Results for Write, Read, and Hold in NV-SRAM.

For read operations, the read signal activates the sense amplifier, which detects small voltage differences on the bit lines. The amplified output is shown as "blout_mem". The verification signals "read_mem" and "blout_mem" confirm that the sensed output correctly matches the stored data. The data line links to BL/BLB and tracks the input/output behavior.

In short, this waveform captures the entire functional cycle-write, retention, and read of a 6T NV-SRAM cell with sense amplifier, specifically validating its use in BNN hardware accelerators where fast and energy-efficient binary storage and retrieval are critical [36].

During a write, when $WL = 1$ and $write = 1$, the storage node Q is forced to follow the input:

$$\Delta V_Q(t) = V_{\text{IN}} \left(1 - e^{-t/(R_{\text{on}} C_{\text{node}})}\right)$$

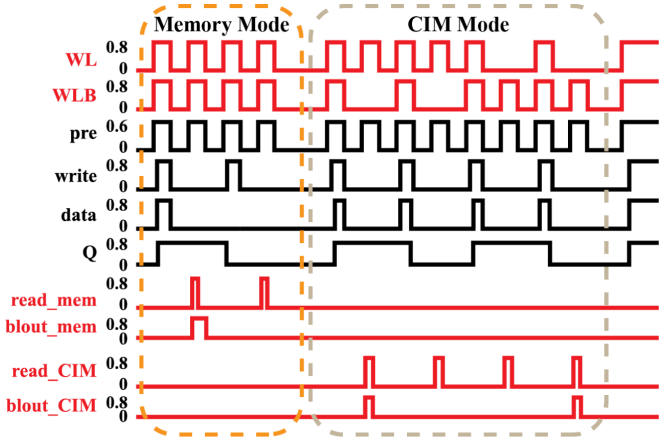


Fig. 12. NV-SRAM Based BNN Architecture in Memory Mode and CIM Mode with Power-on.

The condition for a valid write is:

$$|V_{\text{IN}} - V_Q| > V_{\text{th}}$$

ensuring the cross-coupled inverter pair flips correctly.

During a read, the bit lines are pre-charged to V_{DD}. Once WL is enabled, the cell discharges one bit line:

$$\Delta V_{\text{BL}}(t) = \frac{I_{\text{cell}}}{C_{\text{BL}}} t$$

The sense amplifier resolves when:

$$\Delta V_{\text{BL}} \geq \Delta V_{\text{offset,SA}}$$

The access time is given by:

$$t_{\text{access}} = t_{\text{WL}} + t_{\text{discharge}} + t_{\text{SA}}$$

where $t_{\text{discharge}} \approx \frac{C_{\text{BL}} V_{\text{offset,SA}}}{I_{\text{cell}}}$.

The dynamic power per access follows:

$$P_{\text{dyn}} = \alpha C_{\text{BL}} V_{\text{DD}}^2 f$$

D. Results of NV-SRAM Based BNN Architecture and CIM Mode With Power-Off

The extended transient simulation shown in Fig. 13 validates both NV-SRAM read/write and CIM operations for BNN processing. The access condition of the NV-SRAM cell is defined as:

$$WL = 1 \rightarrow Q \iff BL/BLB$$

During write cycles, the write delay is observed as the time between WL activation and stabilization of Q:

$$t_{\text{write}} = t_{\text{WL}} + t_{\text{d,access}} + t_{\text{inv}}$$

where t_{WL} is WL rise delay, $t_{(d,access)}$ is access transistor conduction delay, and t_{inv} is inverter switching delay.

For reading cycles, the bit line develops a small voltage difference:

$$\Delta V_{\text{BL}}(t) = \frac{I_{\text{cell}}}{C_{\text{BL}}} t$$

With I_{cell} the pull-down current and C_{BL} the bit line capacitance. The sense amplifier resolves the data when:

$$\Delta V_{\text{BL}} \geq \Delta V_{\text{min}}$$

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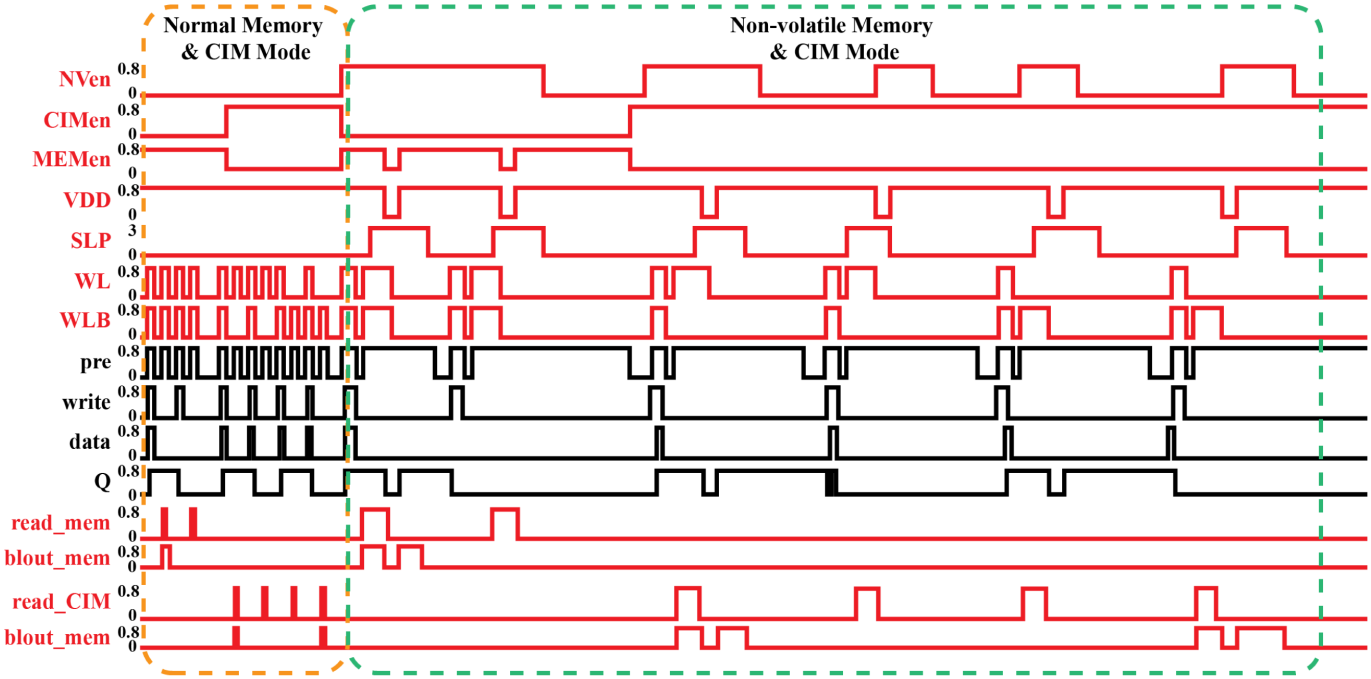


Fig. 13. Hardware Implementation of NV-SRAM Based BNN Architecture.

TABLE III
SUMMARIZED PERFORMANCE PARAMETERS

Parameter	Symbol	Value
Storage Node Voltage	V_Q	0.8 V
Write Delay	t_{write}	∼30 ps
Read Delay	t_{read}	50∼60 ps
Access Time	t_{access}	∼80 ps
Dynamic Power per Access	P_{dyn}	∼5 uW

Yielding a read delay of:

$$t_{\text{read}} \approx \frac{C_{\text{BL}}\,\Delta V_{\text{min}}}{I_{\text{cell}}}\text{ps}$$

In CIM mode, triggered by read_CIM, the bit lines act as accumulators of binary operations. For a set of n cells contributing currents, the bitline develops:

$$\Delta V_{\text{BL,CIM}}(t) = \frac{\sum_{i=1}^{n} I_{\text{cell},i}}{C_{\text{BL}}}\,t$$

The sense amplifier then performs a binary *XNOR* + *popcount* equivalent, where:

$$y = \sum_{i=1}^{n} (x_i \odot w_i)$$

where $x_i \in \{0,1\}$ is the input activation, and $w_i \in \{0,1\}$ is the stored weight.

Thus, the waveform confirms: (1) Stable write; (2) Reliable read; and (3) CIM accumulation of currents across multiple cells implementing XNOR-based BNN operations. Table III summarizes this NV-SRAM Baseed BNN's performance parameters.

V. COMPARISON WITH STATE-OF-THE-ART

As compared with State-of-The-Art, the proposed BNN accelerator integrates non-volatility and CIM capability simultaneously for a standard 6T SRAM, which however has not been established in existing NV-SRAM based systems. The previous SRAM enabled CIM designs are predominantly built upon volatile 6T/8T/10T cells, which suffer from off-chip weight reload after power switching and thus exhibit both high dynamic/standby energy consumption and security threats. On the other hand, existing NV-SRAM architectures are confined to data retention and are not tailored for in-memory XNOR/pop-count computation. The proposed architecture is the first to address this gap by providing persistent weight retention and BNN computation within 6T NV-SRAM at an ultra-low energy (5 fJ/op), compact cell area (7.44 um^2) as well as CMOS compatibility and scalability in 22 nm FDSOI Technology. This NV-CIM co-design ability introduces an entirely new design frame, which balances power with security, persistency and computation efficiency at the same time.

A. Comparison for NV-SRAM Cell

One previous NV-SRAM design is based on ferroelectric capacitor (FeCAP) to back up and restore data, which enable the instant-on feature among other sleep refreshes [15]. It presents one of the initial FeCAP-based NV-SRAM cells in 0.25 um technology at 2.5 V supply with 1.2 ns slower write and read delays (Table IV [15]). Recent NV-SRAM research has concentrated on CMOS-compatible non-volatile technologies such as RRAM aided and spin-based designs. For instance, the following 7T1R RRAM-based NV-SRAM for low-power application in a 16 nm CMOS Technology works at 0.9 V and with write/read delays of ∼52.20/62.63 ps [16]).

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TABLE IV
COMPARISON OF STATE-OF-THE-ART NV-SRAM WITH PROPOSED 6T NV-SRAM

Work	NV Device Type	Tech Node	Supply Voltage (V)	Write Delay	Read Delay
[15]	FeCAP-based NV-SRAM	0.25 um	2.5	1.2 ns slower than standard SRAM	∼
[16]	RRAM 7T1R NV-SRAM	16 nm	0.9	52.20 ps	62.63 ps
[38]	MTJ based NV-SRAM	40 nm	∼	25.33–34.63 ps	∼
[17]	RRAM 8T2R NV-SRAM	16 nm	0.9	53.16 ps	87.98 ps
[18]	SOT-MTJ 7T NV-SRAM	32 nm	∼	2 ns	∼
[19]	RRAM 7T2R NV-SRAM	16 nm	0.9	52.19 ps	86.52 ps
This Work	Capacitor-Coupled 6T NV-SRAM	22 nm FDSOI	0.8	∼30 ps	50∼60 ps

TABLE V
PERFORMANCE COMPARISON OF PROPOSED NV-6T SRAM BNN WITH RECENT SRAM-CIM ACCELERATORS

Work	**Tech / Bit-cell**	**Technology Node**	**VDD (V)**	**Accuracy [%] (MNIST)**	**Throughput (GOPS)**	**Energy Eff. (TOPS/W)**	**Algorithm**	**NV + CIM**
[39]	10T SRAM	65 nm	0.5–1.0	96.62	468	87	Multilayer Perceptron (MLP)	✗
[40]	12T SRAM	65 nm	0.6–1.0	98	5461	139–403	MLP	✗
[41]	8T SRAM	7 nm	1	96.76	455.1	189.3	Multiply-Accumulate (MAC)	✗
[42]	10T SRAM	65 nm	∼1.0	97.12	54.6	273	MAC	✗
[20]	6T SRAM	65 nm	∼	96	18	144	Generative Adversarial Network (GAN)	✗
[21]	12T SRAM	40 nm	0.9	95.90	537	∼	Binary-MAC	✗
This Work	**NV-6T SRAM**	**22 nm**	**0.8**	**96.85**	**8000**	**200**	**MLP**	✓

Also, capacitor-based 8T attack-tolerant NV-SRAM in 40 nm attains 25.33-34.63 ps write delays (Table IV [38]). RRAM based 8T2R NV-SRAM designs in 16 nm and SOT-MTJ based 7T NV-SRAM in 32 nm, however they feature highest write delays of 53.16 ps and 2 ns respectively compared to others (Table IV [17] [18]). 7T2R RRAM-based NV-SRAM for IoT in 16 nm also claims higher write/read delays of 52.19/86.52 ps (Table IV [19]). Therefore, as compared with the above work, Table IV summarizes our proposed capacitor-coupled 6T NV-SRAM in the 22 nm FDSOI Technology. It has all significant improvements to operate at a lower supply voltage of 0.8 V with as low as ∼30 ps write delay, 50∼60 ps read delay, and ∼80 ps access time, while reducing the energy consumption 5 fJ/op, and cell area size to 7.44 um^2.

B. Comparison in System Level

A comparison of proposed BNN accelerator against recent state-of-the-art volatile SRAM-CIM and time domain CIM accelerators is presented in Table V. The accuracy of the proposed design is 96.85% at VDD of 0.8 V, which is competitive with previous works that operate typically at higher supply voltages. While there are some volatile CIM designs demonstrating slightly higher accuracy, they use larger bit-cells and ADC peripherals, but consume large standby power that introduce significant overhead for the entire system [39], [40], [41], [42].

In terms of throughput, our proposed accelerator reaches 8000 GOPS, being as high as the most area efficient time-domain SRAM-CIM BNN implementations and still competitive with charge-domain CIM designs, while also allowing non-volatile computing to be maintained. Meanwhile, our proposed design achieves a energy efficiency of 200 TOPS/W which is similar to several other most energy-efficient SRAM-based CIM BNN accelerators. Unlike prior CIM designs based on volatile memory, our proposed architecture enables true zero-standby-power operation, instant-on inference, and complete elimination of power-cycling traces, which are critical for intermittently powered IoT endpoints, battery-less edge AI, and secure applications. This unique combination of non-volatility, high throughput, strong energy efficiency, and seamless compatibility with standard 6T SRAM cells has not been reported in any existing SRAM-CIM BNN accelerator.

VI. CONCLUSION

This work demonstrates a CMOS-compatible 6T NV-SRAM CIM accelerator for BNNs that uniquely combines non-volatility and high-efficiency in-memory XNOR–popcount computation in a single compact bit-cell. The proposed accelerator with energy per operation as 5 fJ/op and cell area as 7.44 um^2 is implemented. The accelerator is able to store 82000 binary weights in power efficient 10 KB NV-SRAM, and achieves accuracy of 96.85% on MNIST dataset. In summary, this NV-SRAM-based BNN accelerator introduces a new design frame beyond traditional 8T/10T SRAM, SRAM+ADC CIM, and emerging NVM+SRAM CIM architectures, well-fitting to IoT, robotics, autonomous sensors, and embedded vision, in Edge AI applications.

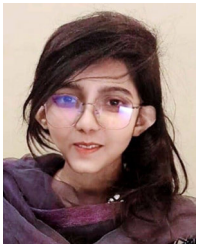
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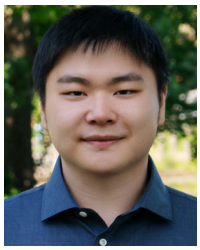
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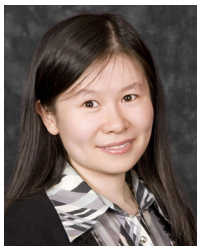
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